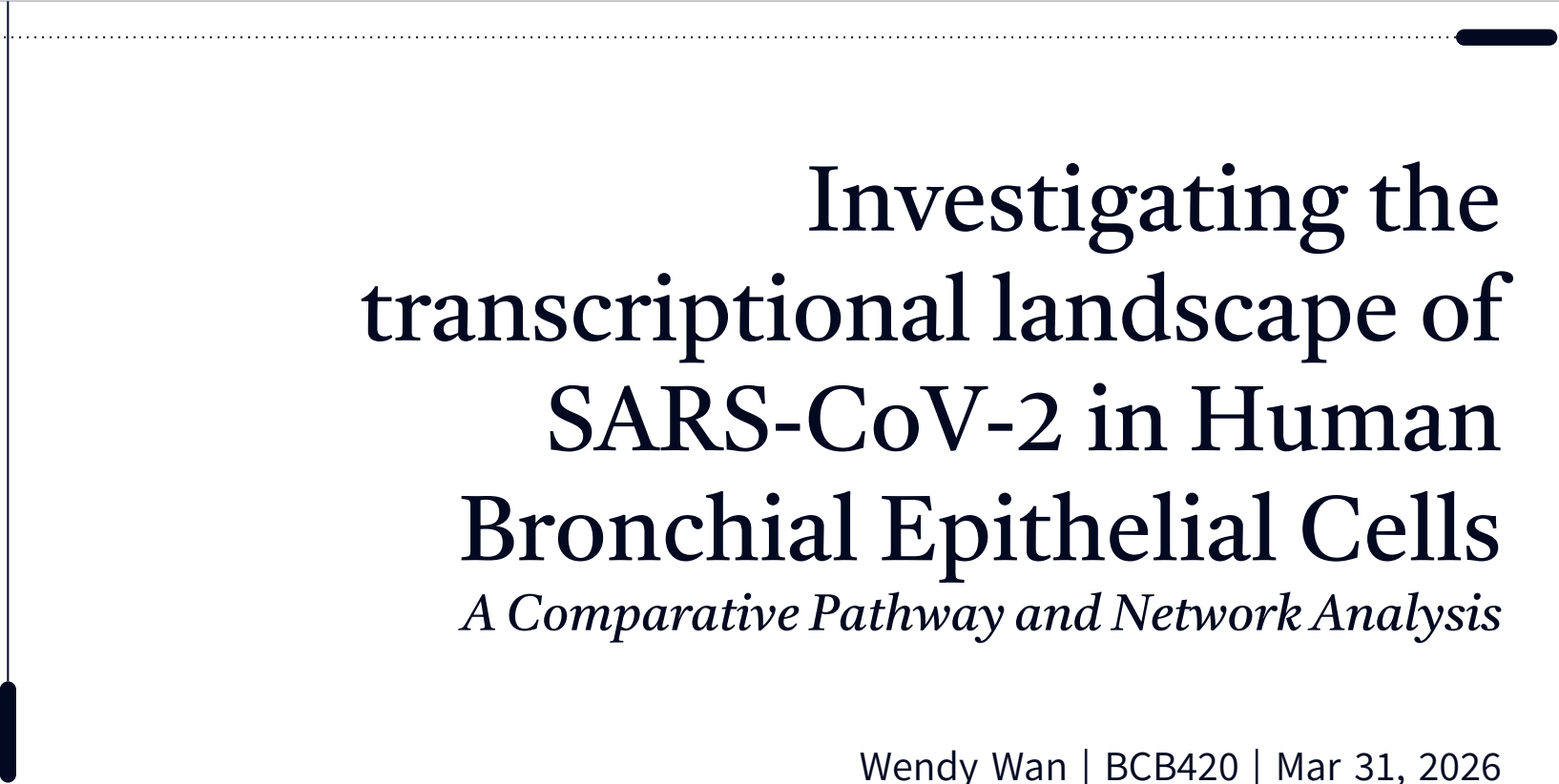




Investigating the transcriptional landscape of SARS-CoV-2 in Human Bronchial Epithelial Cells

A Comparative Pathway and Network Analysis

Wendy Wan | BCB420 | Mar 31, 2026

Motivation

- COVID-19 global pandemic disrupted our daily lives, and permanently altered societal structures worldwide.
- COVID-19 is an infectious disease caused by the SARS-CoV-2 virus.
- Blanco-Melo et al. (2020) compared the transcriptional response of SARS-CoV-2 with other respiratory viruses. Describes the imbalanced host response to SARS-CoV-2 drives the development of COVID-19.

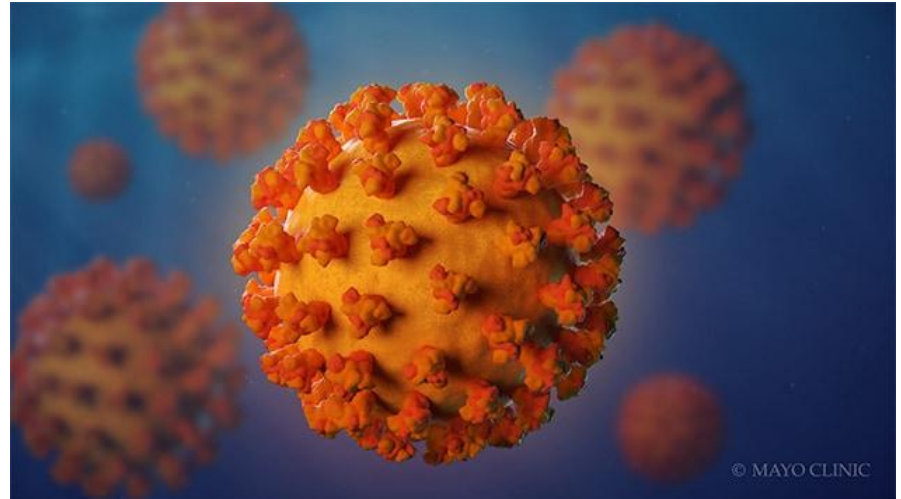


Image from Mayo Clinic



Research Question

How is the imbalanced host
response to SARS-CoV-2 reflected
in the transcriptomic landscape?



Dataset

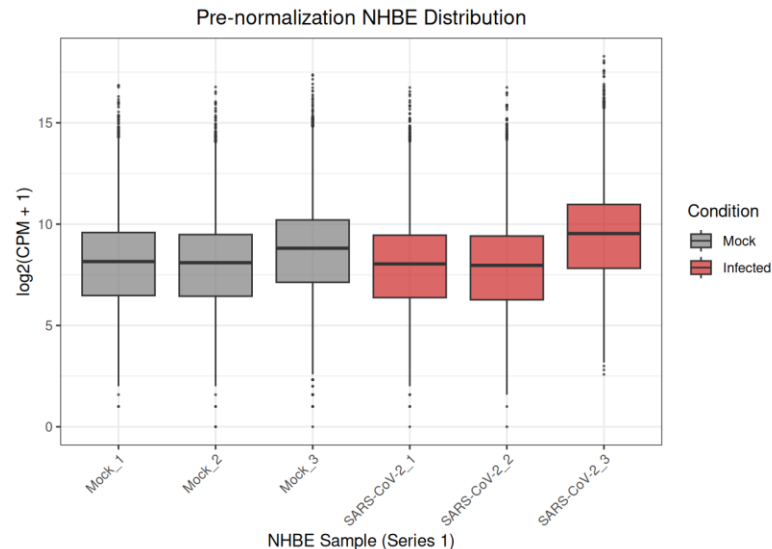
GSE147507

- 78 Samples
- Multiple cell lines (ie. A549 and Calu-3)
- Human, Ferret
- Blanco-Melo et al. (2020)

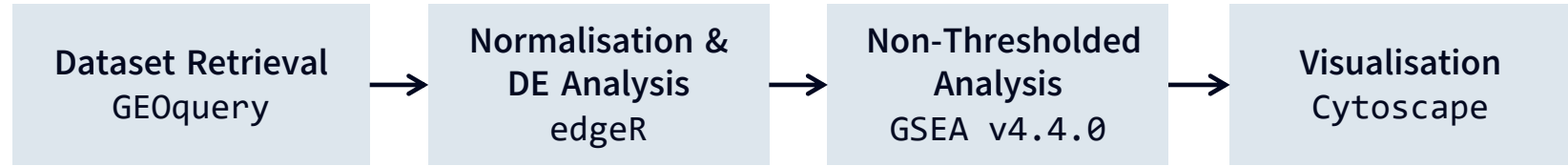
Normal Human Bronchial Epithelial (NHBE) cell line: Primary cells isolated from human airway

6 Samples, 2 Conditions

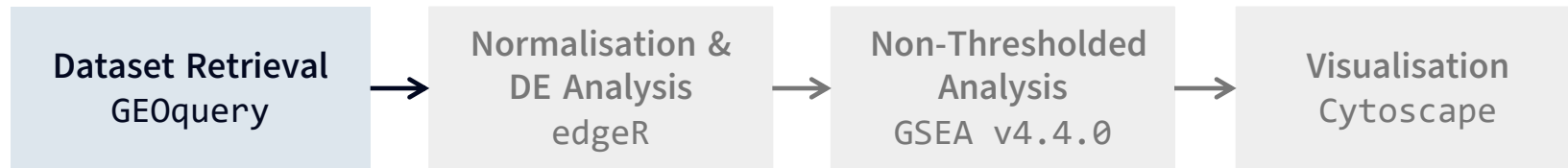
- 3 Mock (Control)
- 3 Infected (SARS-CoV-2)



Workflow



Workflow



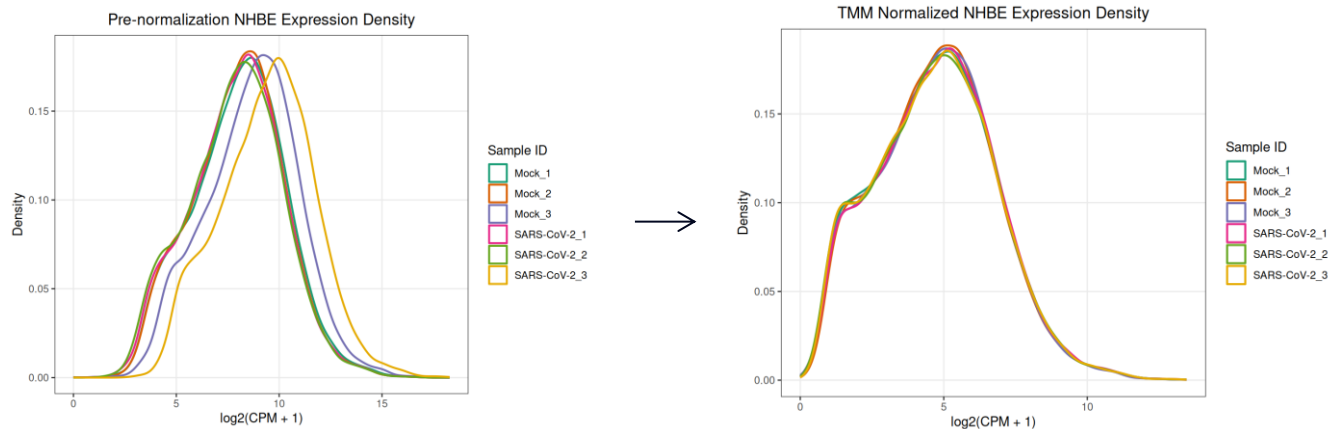
Remove Low Read Counts

- `filterByExpr` from `edgeR`
- Three-sample threshold
- 21,797 → 12,039 genes

Workflow



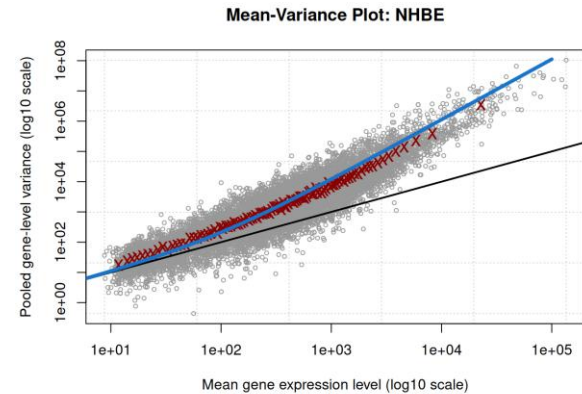
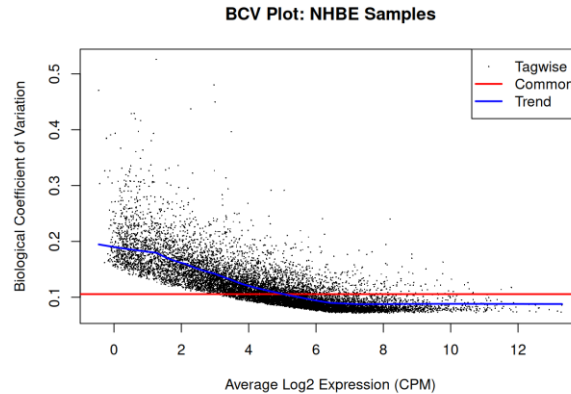
Trimmed Mean of M-values
(TMM)



Workflow



Check for variance and
negative binomial
distribution relationship



Workflow

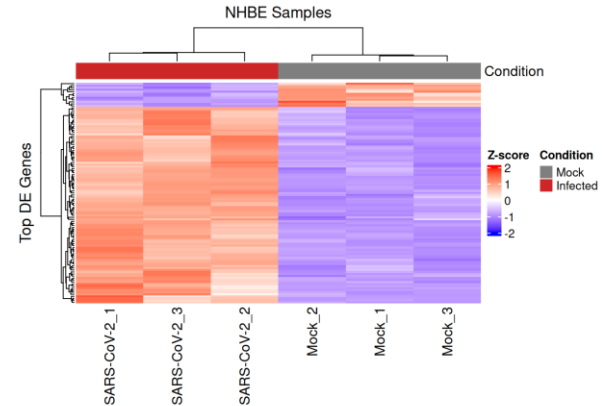
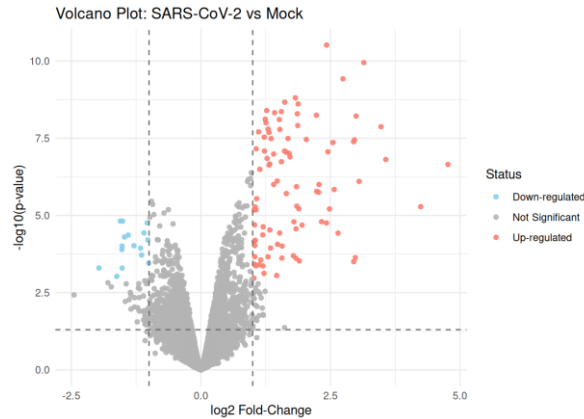


Identify differentially expressed genes

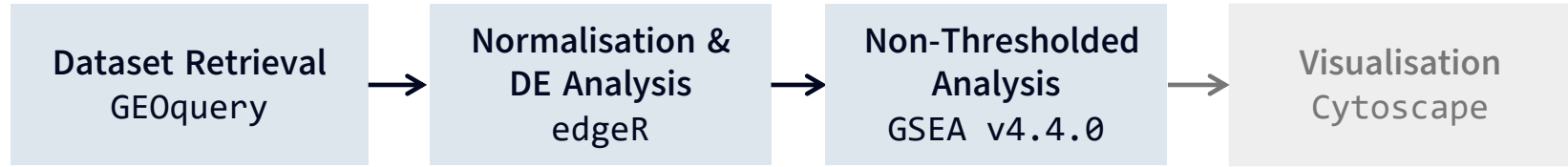
- Quasi-Likelihood (QL) F-test
- Correct p-values using the Benjamini-Hochberg (FDR)

106 differentially expressed genes

- $FDR < 0.05$, $|\log_2FC| > 1$
- 93 upregulated
- 16 downregulated



Workflow



Gene Set Database: BaderLab
Gene Sets (September 01,
2025)

- Gene Ontology Biological Processes (GO:BP),
Reactome, WikiPathways
- No electronic annotations

Rank Score:

- $\log_{10}(PValue) * \text{sign}(\log FC)$
- 12,039 genes

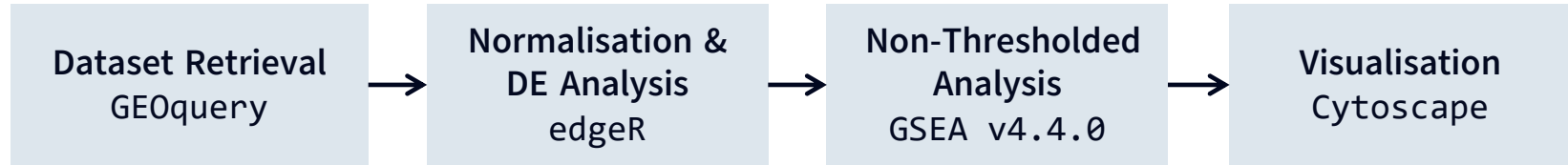
Permutations: 2,000

Size limits: 15-200 genes/set

#	Gene set	Size	NES	FDR q-val
1	HALLMARK_TNFA_SIGNALING_VIA_NFKB	147	2.94	0.000
2	HALLMARK_INTERFERON_GAMMA_RESPONSE	135	2.80	0.000
3	HALLMARK_INTERFERON_ALPHA_RESPONSE	77	2.73	0.000
4	INTERFERON ALPHA BETA SIGNALING	60	2.62	0.000
5	NETWORK MAP OF SARS COV 2 SIGNALING	136	2.56	0.000

#	Gene set	Size	NES	FDR q-val
1	Fibroblast growth factor receptor signaling pathway	20	-1.91	0.729
2	Negative regulation of mitotic nuclear division	32	-1.88	0.536
3	Negative regulation of chromosome segregation	27	-1.86	0.464
4	Negative regulation of nuclear division	34	-1.86	0.362
5	Central nervous system neuron development	26	-1.86	0.295

Workflow



Enrichment Map

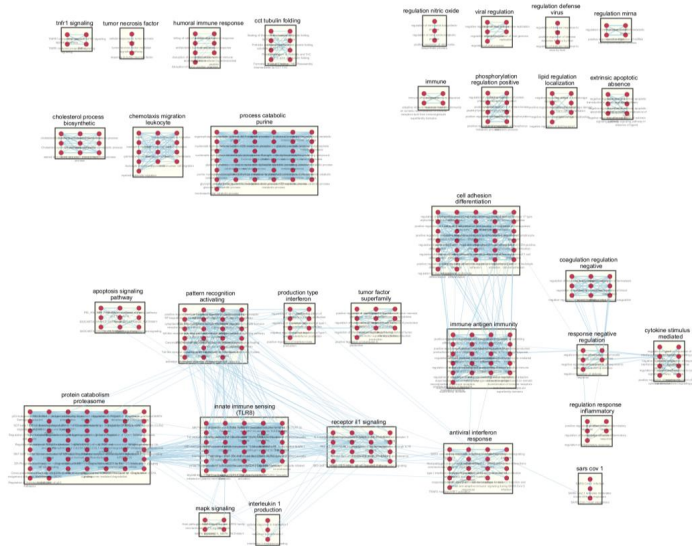
- FDR Q-value cutoff: 0.1
- P-value Cutoff: 1.0
- Edges: Jaccard + Overlap Combined (0.6 cutoff)

Clustering Algorithm: Markov Clustering (MCL)

Labels: WordCloud

- Max words per label: 3
- Min word frequency: 3

745 nodes, 3908 edges



GSEA Results

Top 5 Upregulated Pathways

#	Gene set	Size	NES	FDR q-val
1	HALLMARK_TNFA_SIGNALING_VIA_NFKB	147	2.94	0.000
2	HALLMARK_INTERFERON_GAMMA_RESPONSE	135	2.80	0.000
3	HALLMARK_INTERFERON_ALPHA_RESPONSE	77	2.73	0.000
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5	NETWORK MAP OF SARS COV 2 SIGNALING	136	2.56	0.000

5,213 gene sets

- 3,682 upregulated
- 1,531 downregulated

FDR < 25%

- 806 upregulated
- 14 downregulated

Top 5 Downregulated Pathways

#	Gene set	Size	NES	FDR q-val
1	Fibroblast growth factor receptor signaling pathway	20	-1.91	0.729
2	Negative regulation of mitotic nuclear division	32	-1.88	0.536
3	Negative regulation of chromosome segregation	27	-1.86	0.464
4	Negative regulation of nuclear division	34	-1.86	0.362
5	Central nervous system neuron development	26	-1.86	0.295

Thresholded ORA (gprofiler2)

Top 10 Enriched Pathways for Upregulated Genes

Data sources: GO Biological Process, Reactome, and WikiPathways

- `exclude_iea = TRUE`
- `ordered_query = FALSE`

Significance Threshold:
Benjamini-Hochberg FDR 0.05

Gene set Size: 3-250 genes

Separate Queries:

- 93 up-regulated - 772 significant pathways
- 16 down-regulated - **zero** significant pathways

Source	Term Name	Term Size	Intersection Size	P-value
GO:BP	response to lipopolysaccharide	187	18	2.359e-16
GO:BP	response to molecule of bacterial origin	203	18	9.877e-16
REAC	Interferon alpha/beta signaling	76	13	4.492e-13
REAC	Interleukin-10 signaling	45	11	7.361e-13
GO:BP	cellular response to lipopolysaccharide	142	14	2.56
WP	Overview of proinflammatory and profibrotic mediators	127	15	2.385e-12
GO:BP	defense response to virus	229	16	2.758e-12
GO:BP	cellular response to molecule of bacterial origin	152	14	2.758e-12
WP	Photodynamic therapy induced NF kB survival signaling	35	10	4.460e-12
GO:BP	humoral immune response	249	16	9.403e-12

Themes and Analysis

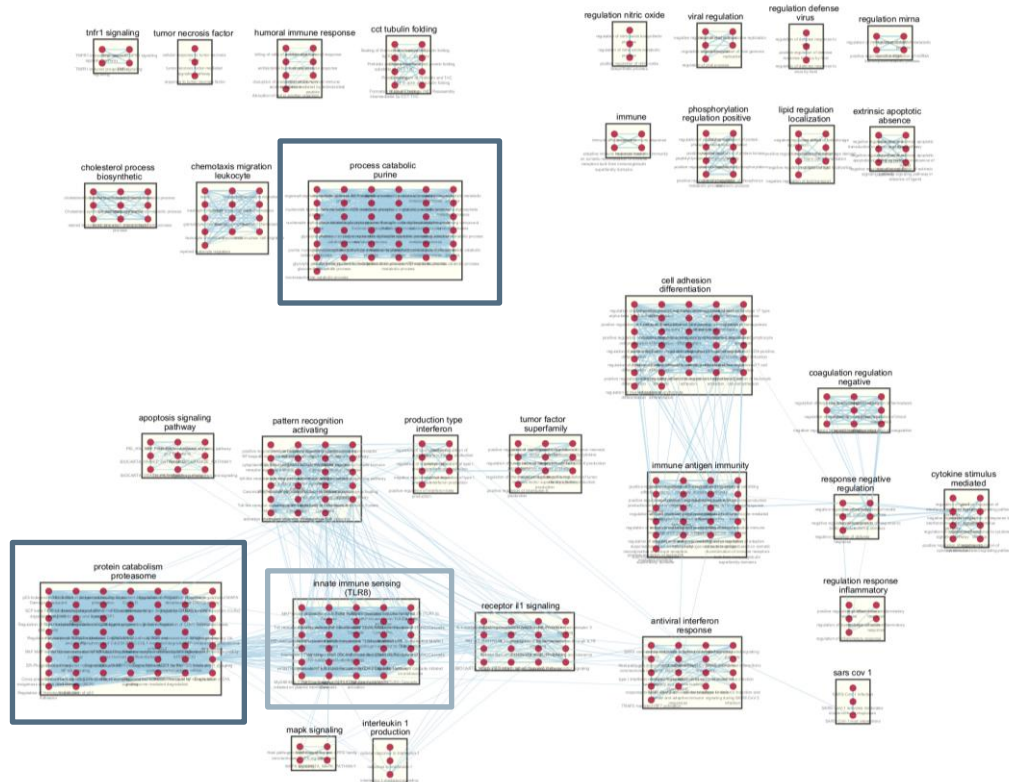
Protein Catabolism & Proteasome

SARS-CoV-2 reprograms host cells, reshaping translation and proteostasis.

- Ubiquitin-proteasome system hijacked for viral replication & assembly (Bojkova et al., 2020).
- Viral protein production causes ER stress and unfolded protein response increasing degradation of misfolded/viral proteins (Gavilán et al., 2024)

Interconnectivity to innate immune sensing cluster (TLR8)

- Adaptor proteins and kinases downstream of TLR8 are activated, modified or turned off for immune related signaling (Cockram et al., 2021)



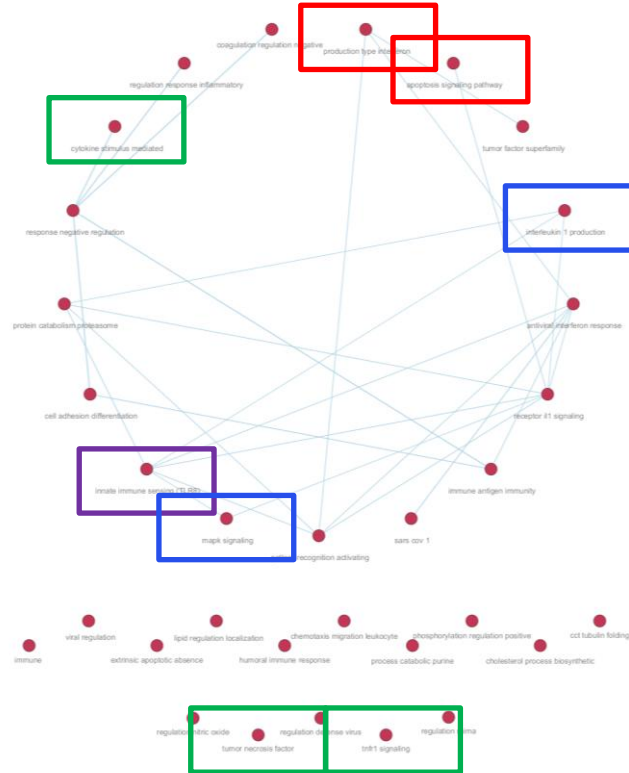
Themes and Analysis

TLR8 Signaling, MAPK Signaling, Interleukin 1 Production, TNFR Signalling and Tumor Necrosis Factor

- **TLR8** Detects SARS-CoV-2 ssRNA, activating **NF- κ B** and **MAPK** pathways, driving transcription of **IL-1 β** , **IL-6**, **IL-8**, **TNF- α** and other **cytokines** that are central to cytokine storm and organ damage (Manik et al., 2021)
- **Cytokine storm**: excessive inflammatory response

Interferon and Apoptosis Signaling Pathways

- Imbalance of low interferons but high cytokines leads to the **cytokine storm** and **tissue damage** (Karki et al., 2022)



References

Blanco-Melo D, Nilsson-Payant BE, Liu WC, Uhl S, Hoagland D, Møller R, Jordan TX, Oishi K, Panis M, Sachs D, Wang TT, Schwartz RE, Lim JK, Albrecht RA, tenOever BR. Imbalanced Host Response to SARS-CoV-2 Drives Development of COVID-19. *Cell*. 2020 May 28;181(5):1036-1045.e9. doi: 10.1016/j.cell.2020.04.026. Epub 2020 May 15. PMID: 32416070; PMCID: PMC7227586.

Bojkova, D., Klann, K., Koch, B., Widera, M., Krause, D., Ciesek, S., Cinatl, J., & Münch, C. (2020). Proteomics of SARS-CoV-2-infected host cells reveals therapy targets. *Nature*, 583, 469 - 472. <https://doi.org/10.1038/s41586-020-2332-7>.

Cockram, P.E., Kist, M., Prakash, S. *et al.* Ubiquitination in the regulation of inflammatory cell death and cancer. *Cell Death Differ* 28, 591–605 (2021). <https://doi.org/10.1038/s41418-020-00708-5>.

Gavilán, E., Medina-Guzman, R., Bahatyrevich-Kharitonik, B., & Ruano, D. (2024). Protein Quality Control Systems and ER Stress as Key Players in SARS-CoV-2-Induced Neurodegeneration. *Cells*, 13. <https://doi.org/10.3390/cells13020123>.

Karki, R., & Kanneganti, T. (2022). Innate immunity, cytokine storm, and inflammatory cell death in COVID-19. *Journal of Translational Medicine*, 20. <https://doi.org/10.1186/s12967-022-03767-z>.

Manik, M., & Singh, R. (2021). Role of toll-like receptors in modulation of cytokine storm signaling in SARS-CoV-2-induced COVID-19. *Journal of Medical Virology*, 94, 869 - 877. <https://doi.org/10.1002/jmv.27405>.